Rendering Slippage in Virtual Reality

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https://github.com/DiarKarim/H2B_MotionBridge/tree/rahaul

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Abstract

The integration of haptic feedback into virtual reality (VR) environments offers promising advances in human-computer interaction, particularly for tasks that require fine motor control and tactile awareness. This study investigates the role of haptic feedback in rendering slippage sensations during object manipulation in VR. A custom experimental setup was developed using a Meta Quest 3 headset and four fingertip-mounted haptic motors controlled via an Arduino-based system. Participants were asked to grasp, lift, and hold virtual objects under two conditions: with and without haptic feedback. During each test, simulated slippage was introduced via directional force signals and the motors were actuated accordingly to mimic tactile sensations. Performance metrics such as task success rate, completion time, and object stability were recorded and analyzed. The results aim to reveal whether haptic feedback enhances the user's ability to perceive and respond to slippage, thus improving precision and control. The findings of this study are expected to contribute to the design of more intuitive and immersive VR systems for applications in teleoperation, robotic surgery, and virtual training environments.

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Contents

List of Figures

1 Introduction

In this section of the paper, background context and motivation are given as to why this research has been done. A problem statement is also given alongside a literature review to summarize key studies in the field. After this, the aim and objective of the paper is given and then the significance and scope of this paper. Finally, a prediction is made on what results and findings will be obtained.

1.1 Background

This paper seeks to investigate the effect of haptic feedback on a user's ability to grasp objects while in a virtual environment. Haptic feedback represents a key advancement in the development of virtual reality as it not only reduces the cognitive load of users, but also provides a higher sensory range compared to only visual feedback experienced within a virtual reality environment.

Despite advances in robotic technology, human-robot interaction has been limited to the use of joysticks and controllers. Operators are constrained to the 3 to 5 degrees of freedom provided by these controllers and joysticks and are unable to make full use of the robot, as some may have up to 7 degrees of freedom. Incorporating VR and haptic feedback would be a bridge for that gap, leading to a higher level of control for the operators/users of these robots, as well as a lower cognitive load.

1.2 Aim

The aim of this paper is to investigate slippage rendering in virtual reality in order to determine whether it can bridge the gap between human-robot interaction by leveraging the immersive capabilities of virtual reality combined with haptic feedback technologies that are used to provide a higher level of sensitivity, leading to higher levels of control. Specifically, this research focuses on rendering realistic slippage sensations to enhance the user's perception of object handling and manipulation within a virtual environment. By improving the fidelity of haptic interaction, the study seeks to reduce the cognitive load on users and enable more natural, intuitive control mechanisms, particularly in scenarios that involve complex tasks such as remote operation, robotic surgery, or remote manipulation. This paper also aims to evaluate the effectiveness of various haptic feedback approaches in conveying slippage and to assess their impact on user performance, perception, and task accuracy in simulated environments.

1.3 Significance & Scope

This research is limited to investigating haptic feedback delivered through the fingertips of the right hand within an immersive virtual reality environment that was created in Unity. The experimental task involves grasping, lifting and holding an object for a specified duration across repeated trials. Participants perform the task under two different conditions: with and without haptic feedback, allowing for a direct comparison of performance and perception influenced by tactile input.

1.4 Predictions

Participants who receive haptic feedback through the haptic system placed on their fingertips are predicted to demonstrate better performance in the grasping and regrasping tasks compared to those who do not receive any haptic feedback. Specifically, it is expected that they will exhibit a higher variation in angle changes in the direction from which the force may have come from. Participants are also predicted to score lower on the NASA TLX load index in haptic feedback trials compared to those without haptic feedback. This is because haptic feedback would reduce the cognitive load produced from these trials and allow for a more specific response from the participants towards the direction the force originated from.

2 Literature Review

2.1 Haptic Feedback in VR

Haptic feedback is increasingly regarded a key technology in enabling virtual reality (VR) systems to become more immersive, intuitive, and effective, particularly in scenarios involving fine motor control and realistic object interaction. By introducing tactile sensations that emulate the perception of touch, pressure, and force, haptic systems help bridge the sensory and perceptual gap between virtual and physical environments. These systems are especially beneficial for tasks that require precision, such as grasping, maintaining object stability, and executing coordinated hand movements. Such tasks form the foundation of slippage rendering techniques, which are the primary focus of this study.

Early explorations into the application of haptic feedback in VR were led by the medical field, where the realism of virtual training environments is important. In a foundational study, van der Meijden and Schijven evaluated the efficacy of haptic feedback in both traditional and robot-assisted minimally invasive surgical simulations [1]. They concluded that the inclusion of haptic elements improved user performance, improved the precision of technical maneuvers, and accelerated the pace at which surgical skills were acquired. The benefits were largely attributed to the ability of haptic feedback to convey subtle sensory information, such as small changes in resistance or tissue compliance. This level of detail is essential in contexts like slippage rendering, where perception must be sensitive to variations in surface interaction and object behavior. In such cases, haptic feedback must be both highly accurate and dynamically responsive to simulate pre-slip and slip events.

The capacity of users to interpret haptic cues in VR was further explored by VåPenstad et al., who demonstrated that high-fidelity tactile signals could be effectively interpreted by users when presented with sufficiently responsive systems [2]. Their findings reinforce the importance of the resolution of the haptic system and the temporal responsiveness in rendering subtle physical interactions. In the case of slippage, where the user must detect small changes in position or surface friction, these capabilities become essential to convey accuracy and enable timely corrective responses.

In addition to this, Otaduy et al. [3] and Stone [4] examined how force feedback systems could be tuned to enhance realism in virtual environments, particularly by simulating continuous low-amplitude lateral movements typical of object slippage. Their insights underline the importance of modeling dynamic interaction forces in real-time to preserve the illusion of physicality. In line with this, Yang et al. [5] emphasized that rendering tactile phenomena such as compliance, slip, and surface texture is instrumental not only for realism but also for fostering user trust and enhancing the intuitiveness of object manipulation in VR.

Technological advances have further enabled the miniaturization and sophistication of haptic actuators, making wearable tactile systems more viable and expressive. For example, Shi and Shen introduced a wearable soft-material device capable of providing combined thermal and tactile stimuli [6]. This multisensory approach increases the fidelity of the virtual interaction by closely mirroring real-world feedback. Jacucci et al. [7] extended this conversation by exploring the dimensions of touch, hypothesizing that even minimal tactile cues, such as vibrations mimicking slippage, can carry contextual and significance in virtual reality interactions.

The potential applications of high-resolution haptics extend beyond the fields of industrial or medical training. Yong [8] and Tang [9] illustrated how haptic-enhanced VR can transform interactive gaming and training environments, especially those requiring responsive feedback to rapid changes in contact dynamics. This aligns with the goals of this paper which focuses on investigating slippage rendering, where moment-to-moment adjustments in finger pressure and object behavior are essential. Similarly, Kovanur Sampath et al. [10] investigated the integration of tactile feedback into wearable technologies, highlighting their role in enabling precise object manipulation and promoting sensory-motor learning. Their work supports the argument that interfaces that support tactile enhancements are fundamental for the development of realistic VR systems capable of conveying subtle physical transitions, such as those encountered during slippage events.

Despite the growing complexity of these systems, simulating slippage remains uniquely challenging due to its reliance on temporary and multidirectional physical cues. Effective slippage rendering requires accurate real-time sensing of object dynamics, advanced friction modeling, and the seamless transmission of continuously variable feedback to the user. As noted by Gibbs [11], achieving perceptual realism in VR requires a delicate balance between system responsiveness, hardware constraints, and user cognitive load.

In contexts where hardware cannot deliver true mechanical feedback, perceptual illusions must be carefully designed to substitute for tactile experiences without introducing dissonance.

From a foundational perspective, Aziz [12] laid conceptual groundwork in the area of sensory substitution, which remains highly relevant to the challenge of replicating slippage. Vibrotactile feedback, though abstracted from direct mechanical simulation, can be employed to convey a meaningful perception of object motion or texture change, especially when timed and scaled correctly in response to user actions.

In recent years, the conversation has also expanded to consider ergonomics, usability, and system integration. As VR systems become more portable and accessible, the demand for lightweight, low-power, and skin-compatible haptic solutions has grown. Research now includes investigations into how wearable systems can provide bidirectional feedback or adapt feedback patterns based on user behavior and context. These developments are particularly promising for simulating slippage, which is not merely a function of force but of tactile timing, directionality, and user awareness.

As haptic feedback technologies continue to evolve—offering improved spatial resolution, faster signal transmission, and multimodal feedback integration, the vision of fully immersive, responsive virtual environments capable of convincingly simulating slippage becomes increasingly realistic. Understanding both the foundational and emerging research in this field is critical for advancing the design of next-generation virtual reality systems. The studies reviewed in this section collectively form a theoretical and technical preparation for future developments in haptic-enhanced virtual manipulation, where users can not only see and move objects but genuinely feel and interact with them as if they were physically present.

2.2 Grasping in VR

Grasping in virtual reality (VR) is a central challenge in developing systems that offer realistic and immersive user experiences. Unlike traditional interfaces that rely on joysticks or button-based input, natural hand-based interaction demands accurate tracking, complex gesture recognition, and, most importantly, tactile feedback that conveys the physicality of virtual objects. High-fidelity grasping simulation must account for force application, contact stiffness, object compliance, and slippage behavior, all while maintaining responsiveness and perceptual coherence.

Blaga et al. [13] proposed VRGrasp, a novel taxonomy that formalizes the various types of human grasp observed in virtual object interaction. This taxonomy serves as a foundational framework for building grasp-aware systems capable of translating natural hand poses into meaningful virtual manipulations. Zhou [14] complemented this work by identifying the importance of adaptive contact modeling in real-time applications, highlighting the need for simulation environments to respond to dynamic grip changes and finger pressure changes.

The social dimension of grasping was explored by Shi et al. [15], who introduced a multi-user virtual reality environment where the manipulation of shared objects enabled collaborative tasks. Their findings underscore the necessity for grasping interfaces to adapt to distributed force contributions and spatial negotiation between users. Such collaborative frameworks are particularly relevant in domains like rehabilitation, virtual co-design, and telepresence, where synchronous and nuanced manipulation is vital.

From a hardware development point of view, Liu et al. [16] and Chessa et al. [17] contributed significantly to wearable haptics by designing systems that can simulate realistic physical resistance, friction feedback, and modulation of grip force. These devices enable users to experience tactile qualities, such as surface texture and object mass, bridging the sensory divide between virtual and real interactions. The integration of these wearables with precise hand tracking paves the way for naturalistic and immersive manipulation experiences.

Historically, Borst and Indugula [18] were among the first to develop grasp-aware physical models that simulate object deformation and force reactions in response to user input. Their work laid the groundwork for engines capable of modifying object behavior based on real-time grip strength and finger placement, essential for tasks such as secure holding, dynamic reorientation, and slippage control.

Visual feedback mechanisms have also been shown to be effective in compensating for the absence of physical haptics. Geiger et al. [19] and Canales et al. [20] demonstrated that realistic rendering of hand-object contact, when paired with subtle visual cues such as object displacement or finger indentation, can create a convincing illusion of touch. These techniques are particularly advantageous for consumer-grade VR systems, where full-scale haptic integration may be cost- or hardware-prohibitive.

Beyond conventional approaches, Arata and Jingu [21] proposed a double-sided tactile stimulation system that simulates force transmission across the palmar and dorsal surfaces of the hand. This design is particularly promising for conveying balance shifts, dynamic weight changes, or early slippage detection, where multidirectional cues can enhance realism.

Precision interaction tools were explored by Thibault-Delrieu [22], who designed fine motor control instruments suitable for surgery simulation and micro-assembly training. These tools operate under low-latency, high-resolution feedback constraints and underscore the demand for extremely responsive systems in high-stakes training applications.

Gesture-based interaction models have also gained attention, with Fernandez and Oprea [23, 24] comparing natural hand gesture systems to traditional controller input. Their studies show that while gesture systems are often perceived as more intuitive, current limitations in hand-tracking fidelity and environmental occlusion affect performance accuracy, especially during object transitions or complex grip configurations.

To address the limitations of hardware-based haptics, Mores Prachyabrued and Borst [25] advanced visual-tactile substitution strategies. By aligning force-based visual deformations with expected haptic outcomes, they showed that users can be led to perceive tactile properties without direct touch, an approach that supports low-budget or mobile VR platforms.

Levin [26] explored the underlying quality metrics of interactions between objects and the hand, identifying key factors such as interface stability, grasp retention, and tactile consistency as determinants of the success of manipulation. These insights directly support the argument for incorporating high-quality force feedback and physical simulation in VR grasping pipelines.

Looking ahead, the convergence of precise hand modeling, physically adaptive object behavior, and multimodal feedback (e.g., visual, haptic, and auditory) is critical for pushing the boundaries of realism in VR grasping. With ongoing improvements in haptic hardware, computer vision, and biomechanics-informed modeling, we can expect future systems to provide users with the dexterity and responsiveness of real-world interaction. As the fidelity of these systems increases, so will their utility in areas ranging from education and remote training to rehabilitation and industrial prototyping.

2.3 Human-Robot Interaction in VR

Human-robot interaction (HRI) in virtual reality (VR) is a growing area of research that leverages haptic feedback to simulate realistic communication, cooperation and control between humans and robotic agents. As virtual reality becomes a more common interface for robotic research, training, and teleoperation, the importance of haptics in replicating physical interaction and improving task outcomes has become increasingly clear. This section reviews key applications of haptic feedback in HRI-VR systems, highlighting both functional and experiential benefits across domains.

Leeper et al. [27] explored user strategies for physical HRI in virtual environments, demonstrating that haptic-enabled feedback improved both robot manipulation and user confidence. Their study established that incorporating force signals into robot programming tasks helps users better predict robotic responses and develop more accurate mental models. Similarly, Murnane [28] discussed the role of multimodal feedback, including haptics, in supporting collaborative robot learning environments, where users train robots by demonstration.

In the context of social interaction, Wang et al. [29] developed a haptic-enabled handshake simulation between virtual avatars and robotic proxies. Their findings suggest that physical contact enhanced by force and vibration feedback significantly improves the realism and social presence of robotic agents in virtual meetings. Walker [30] extended this idea to educational HRI, where haptic cues were used to enhance robot-student interactions in VR classrooms, enhancing engagement and attentiveness.

Schmitz et al. [31] presented one of the first implementations of robotic rehabilitation using haptically enhanced virtual reality. Their system allowed patients to interact with virtual robotic arms that provided real-time force feedback during therapy exercises. This approach has proven especially effective in restoring motor control and increasing user motivation in neurorehabilitation contexts. Bustamante et al. [32] also contributed to this field with ArmSym, a virtual reality rehabilitation system that facilitates HRI through bilateral robotic interaction, force detection, and motion tracking.

Industrial and collaborative robotic applications have also been integrated with VR-based haptics. Matsas and Vosniakos [33] implemented a maintenance training simulation where users learned to guide and program robotic arms in virtual environments. Through tactile feedback, users developed a better sense of the position of robotic tools and the response to torque. Rückert et al. [34] applied similar principles to emergency robotics, developing a virtual reality interface that allows operators to simulate hazardous scenarios involving autonomous machines, with haptic cues replicating physical resistance and feedback from remote robots.

In advanced robotic manipulation tasks, Adami et al. [35] investigated how haptic guidance enhances human control over robotic arms, particularly in precise teleoperation scenarios. Their findings demonstrated that force signals improve fine motor control and reduce operator fatigue. Wang [36] contributed by developing intelligent haptic systems that adapt force feedback in real time based on user behavior, enabling more seamless and efficient HRI.

Liu et al. [37] explored the HRI user experience within immersive environments, finding that perceived agency and control over robotic avatars in VR are significantly improved when there is tactile feedback. Their findings point to the importance of multimodal cues in building trust and maintaining user attention during extended interactions.

Williams et al. [38] examined collaborative mixed reality interfaces for HRI, highlighting how shared spatial awareness and haptic-assisted coordination improve task efficiency in joint problem solving. Their research illustrates how wearable and environmental feedback systems can enhance human-robot teamwork in immersive training simulations.

From a systems design perspective, integrating haptic feedback into VR-HRI setups improves not just realism but also user trust, error reduction, and task performance. These systems offer an intuitive and immersive platform for training users in complex robotic workflows without requiring direct exposure to physical robots. In addition, they facilitate safe testing environments for human-robot collaboration in high-risk or inaccessible settings.

2.4 Wearable Haptic Technology in VR

Wearable technology has become a cornerstone of innovation in virtual reality (VR), enabling a more natural, embodied interaction with digital environments. These systems, ranging from gloves and exoskeletons to soft materials and textile-integrated sensors, provide real-time feedback to simulate tactile sensations, motion, and resistance. Wearable haptic devices in VR not only enhance immersion, but also improve performance in training, therapy, and entertainment contexts.

Choi et al. developed two influential wearable haptic devices: Grabity [39] and Wolverine [40]. Grabity provides kinesthetic feedback using a motorized slider to simulate weight and inertia during object manipulation, while Wolverine offers directional force feedback through retractable brake mechanisms that restrict finger motion. These innovations allow users to experience virtual object properties more realistically and have paved the way for more advanced hand-mounted interfaces.

Expanding on force feedback design, Hinchet et al. introduced Dextres, a soft wearable haptic device that delivers force signals via electroactive polymers [41]. Unlike rigid devices, Dextres conforms to the anatomy of the hand, offering a lightweight and low-profile solution for immersive interaction. Similarly, Zhu et al. proposed a soft haptic glove using dielectric elastomer actuators to enable bidirectional tactile feedback, making it suitable for continuous interactions in complex VR tasks [42].

Yin et al. [43] explored wrist-wearable systems that simulate directional friction to emulate slippage and object dynamics. Their work demonstrates how compact wearable actuators can effectively convey nuanced motion cues such as drag, inertia, or change in load distribution. These techniques are particularly useful for simulating subtle physical interactions that go beyond binary touch feedback.

From an ergonomics and comfort point of view, Lee et al. [44] reviewed the integration of wearable haptics into VR systems, highlighting challenges such as thermal management, power consumption, and skin irritation. Their insights highlight the importance of designing devices that are both technically robust and comfortable enough for extended use.

Touch perception is also enhanced through vibrotactile and skin-stretch mechanisms. Perret et al. [45] developed a wearable sleeve that simulates contact forces and textures using shape memory alloys. Spagnoletti et al. [46] investigated the rendering of soft object contact through inflatable fingertip actuators, contributing to tactile realism in virtual object exploration.

Frisoli et al. [47] proposed a hybrid glove that combined kinesthetic and vibrotactile elements to simulate a wider range of forces during grasping and manipulation tasks. Their evaluation shows improved user immersion and task performance, particularly in delicate interaction scenarios. In a similar direction, Prattichizzo et al. [48] emphasized the importance of decoupling force and motion feedback channels to avoid perceptual conflicts, improving the clarity of tactile cues.

Maisto et al. [49] performed a comparative evaluation of various portable devices, finding that multimodal systems (those that include kinesthetic and cutaneous feedback) outperform single mode solutions in complex manipulation tasks. Their findings underscore the role of wearable haptics in enabling dexterous and nuanced control of virtual objects.

Lastly, Saint-Aubert [50] contextualized wearable haptics within the wider evolution of virtual reality, emphasizing its transition from assistive technology to mainstream input methodology. As the lines between human and digital embodiment blur, wearable systems will likely serve as foundational interfaces for the next generation of immersive computing.

In summary, wearable technology in virtual reality is rapidly advancing toward systems that are lightweight, multimodal, and user-centric. These devices offer a compelling bridge between digital content and human perception, enabling intuitive and lifelike interaction. The reviewed literature highlights the critical role of wearables in delivering physicality, responsiveness, and realism, key ingredients for immersive and effective virtual experiences.

3 Methodology

This section details the experimental design, equipment setup, system architecture, procedures, and data handling protocols used to evaluate the impact of haptic feedback on slippage detection and perception in virtual reality environments.

3.1 Equipment & System Architecture

The structure of the experiment was based on a Meta Quest 3 virtual reality (VR) headset, which was chosen due to its ability to track 6 degrees of freedom and markerless hand tracking, which is integral to the performance of hand tracking within the Unity environment. The capability to track 6 degrees of freedom allows movements in the x, y and z directions to be tracked, as well as rotation around those three axes (pitch, yaw, roll). The Quest 3 was then connected to a computer that is VR ready using a Quest Link cable, enabling high-fidelity rendering and low-latency interactions throughout the experimentation phase.

The haptic feedback was simulated using four haptic feedback motors, rendered in **Figure**

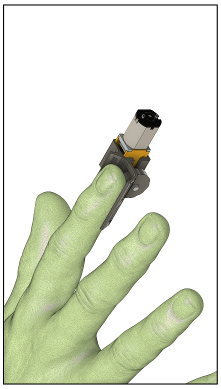

Figure 1: Render of the Motors

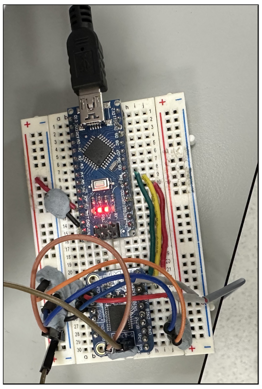

Figure 2: Arduino Breadboard.

Figure 3: Housing for the Arduino Boards.

1, that were connected to the unity application paired with the Meta Quest 3. The haptic feedback motors, which simulated the haptic feedback, consisted of a geared 5V DC motor controlled through a TB86612 motor driver, which was interfaced with an Arduino Nano microcontroller. The microcontrollers were programmed so that they would receive a simple character set command that would allow them to control the motor modes. Here, characters f and b would activate the front and back rotation of motor 1, respectively, while characters g and n were used for motor 2. The command sets were replicated on two Arduino boards as shown in **Figure 2** and **Figure 3**, making it possible to independently drive four haptic actuators.

The Unity engine was configured to issue these commands based on pre-programmed object interactions and slippage events. This was achieved through a custom C# script that would trigger the motors on each finger depending on the direction the force was originating from. For example, if the force originated from the north-west region relative to the user's palm, then motors on the index finger would be triggered.

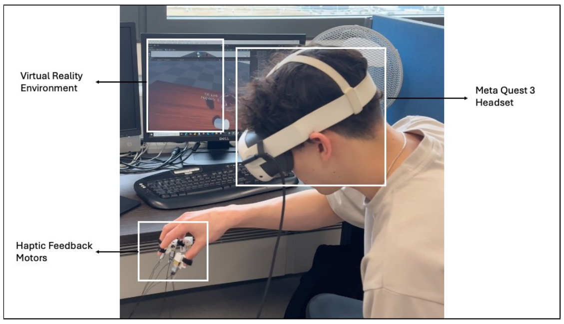

Figure 4: The setup for the user trials

3.2 Experimental Procedure

A user study was conducted to assess the feedback of participants who experienced haptic feedback during simulated slippage. 14 participants within an age range of 18 to 23 were found and each of them was asked to sign a consent form created by the researchers. The participants were then seated and equipped with the Meta Quest 3 headset as seen in **Figure 4** and placed in the virtual environment created in the Unity editor as seen in **Figure 5**, **Figure 6** and **Figure 7**. Prior to the main trials, each participant completed a short orientation phase, where they were informed on what the trials would be about, as well as an explanation on what to expect when in the virtual environment, as well as how the haptic feedback motors would function. The haptic feedback motors were connected to the participants' thumb, index, middle, and ring fingertips and then actuated to familiarize them with the feel of the haptic feedback, as well as the VR environment and grasping mechanics. During this orientation phase, calibration is also performed in order to map the participants' hands to the robot gripper digits, this ensured that the hand alignment was accurate to avoid incorrect slippage and discomfort during the trials.

Figure 5: Grasping in the VR Environment.

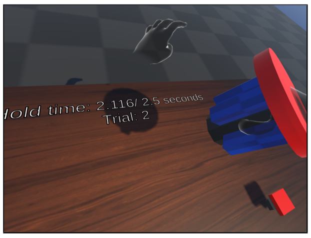

Figure 6: Lifting phase of the trials.

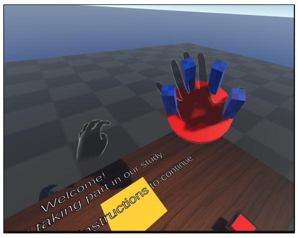

Figure 7: Robot Gripper in the VR Environment.

The four motors mounted on the fingertips of the participants were attached using a 3D printed housing, which holds the motor as seen in **Figure 8**, **Figure 9** and **Figure 10**. These housings then have Velcro straps passed through them in order to secure the housing onto the participants fingertips, these were designed with the intention to maintain stable contact without restricting finger movement.

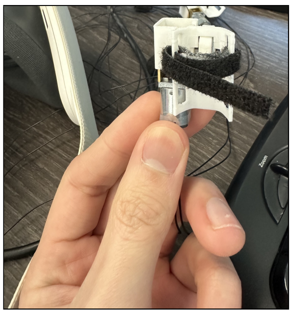

Figure 8: Top of the haptic feedback housing.

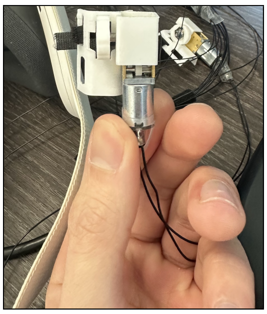

Figure 9: Bottom of the haptic feedback housing.

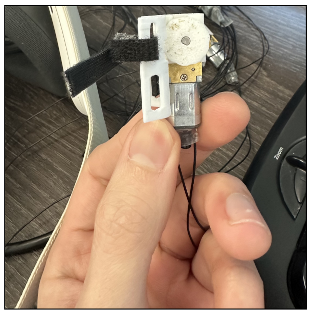

Figure 10: Wheel of the haptic feedback housing.

The calibration was implemented through a mapping function as shown in **Figure 11**. When the key o is pressed on the keyboard, that tells the unity script to register the user's hand at that point in time as the open state for the digits on the robot gripper, and when the key c is pressed on the keyboard, it tells the unity script to register the closed finger state instead. This calibration allows the robot gripper to mimic the participants grasping in real life.

```
// Mapping function to get the value between 0 and 1.
4 references
public static float map(float value, float leftMin, float leftMax, float rightMin, float rightMax)
{
    return rightMin + (value - leftMin) * (rightMax - rightMin) / (leftMax - leftMin);
}
```

Figure 11: Mapping Function

After this orientation phase is completed, participants are then asked to start the trials. The trials consist of the users being placed in 4 blocked trials, each consisting of 5 trials in which they will either receive haptic feedback in the first and third block, or in the second and fourth block. They are then asked to press the button within the virtual environment to instantiate the target object, which is a cylinder. Once this cylinder has been instantiated, participants are then asked to grasp and lift the cylinder above a certain height. Once the object has been raised above that height, a timer on the table within the environment begins counting up to 2.5 seconds, the time needed to hold the object above the mark stays the same for all other trials. After this time has elapsed, a force is applied in one of 7 random directions. These directions are north, north west, west, south west, south east, and north east. Depending on the direction from which the force originates, the haptics are actuated accordingly. Once this force has been applied and the haptics are actuated, the cylinder is dropped from the participants' grasp, and the object is destroyed and reinstantiated to be placed on the table. The participants are then asked to regrasp the object based on the visual and haptic feedback that they experienced during the previous phase of the trial. After regrasping, the target object is then destroyed, and the trial ends, at which point the participants are prompted to press the button on the table again in order to begin the next trial. The participants then repeat 19 more trials that are similar to the initial one except that every trial the direction from which the force originates is randomly chosen from one of the seven directions previously stated.

3.3 Data Collection and Handling

Data from these trials were collected systematically through a dedicated function embedded in the main script of the VR application. This function instantiated data classes to capture relevant variables, including spatial coordinates of the user's hand, as well as the target object, event timestamps, and the trial number and direction of force. Each interaction phase (e.g. object grasp, hold, and slippage) was associated with event tags, allowing for segmented analysis of user performance across the task timeline.

Each participant completed 20 trials, and the corresponding data was saved as individual files labeled per session. Participants were briefed prior to participation and provided their written informed consent. All data collected were anonymous and securely stored. Access to raw data was restricted to members of the research team, according to ethical research guidelines.

After these results were collected and made anonymous, they were uploaded to a Google Drive and used in Google Colab to develop a visualization of the angle changes that we are looking for in order to compare the results.

The NASA TLX Load Index results were uploaded to a Microsoft Excel Sheet in order to create a pivot table comparing the trials of participants with haptics and the trials without the haptics for the same participants.

3.3.1 Ethical Considerations

Prior to the start of the study, ethical approval was obtained according to the university's research ethics guidelines. All participants were fully informed about the nature and scope of the experiment and were required to provide their written informed consent. They were informed of their right to withdraw at any time without consequence.

To protect participant anonymity, no personally identifiable information was recorded during the trials. The data collected was securely stored in anonymized form, with access restricted to the research team. The study design adhered to the principles of respect, autonomy and data confidentiality, ensuring participants' physical and psychological safety throughout the trial.

No adverse events were reported during the study and the use of haptic wearable hardware was tested for comfort and safety during a calibration phase prior to each session.

4 Results and Analysis

This section focuses on presenting the results derived from the user studies that were conducted in order to evaluate the effect that haptic feedback has on a user's ability to regrasp, as well as if there is any effect of a lower cognitive load on users' when assisted with haptic feedback throughout the trials. The collected data includes the angle changes recorded between the lifting phase of the trials and the regrasping phase, along with an overall index score derived from the NASA TLX Load Index.

4.1 Grasping Task Results

In this segment, we analyze the results of the user trials that were conducted. Each participant underwent 20 trials in blocks of 5, each block was haptic or non-haptic. The data recorded was then isolated and an analysis was performed on the specific stages of each trial. The trials are split into events within the code so that the data is more easily readable later on when processing the jSon file. These events are called: 'trialstart', 'lifting', 'holdingabove', 'dropping', 'hapticfeedback','resettarget','regrasping', 'trialend', and 'holdingbelow'.

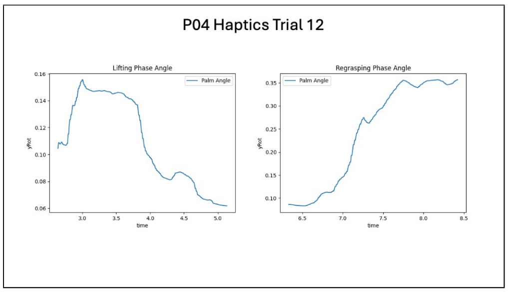

Figure 12: Haptics Trials (radians)

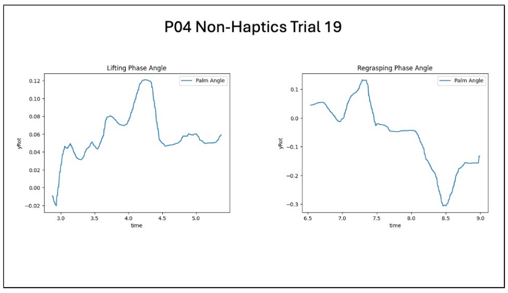

Figure 13: Non-Haptics Trials (radians)

The graphs in **Figure 12** and **Figure 13** represent two separate trials undergone by the same participant. Both trials had the same force direction to the north east. yRot represents the rotation of the participant's hand on the y-axis in radians. We can use this yRot to identify angle changes over time throughout the different events, for example, a positive value of yRot represents a clockwise rotation.

Figure 12 presents changes in the palm angle (yRot) for participant P04 during Trial 12, which included haptic feedback. The graph on the left shows the lifting phase, which began shortly before the 3-second mark. There is a sharp and consistent increase in the angle, peaking at around 3.2 seconds, followed by a gradual decline until approximately 5 seconds. This pattern indicates a distinct hand rotation to achieve better grip at the beginning of the lifting motion, with steady adjustment throughout the event. On the right, the regrasping phase begins just after 6.5 seconds and shows a consistent upward trend in the palm angle. The angle increases substantially until around 7.8 seconds and then levels off, remaining relatively stable through to 8.5 seconds.

Figure 13 displays data from participant P04 during Trial 19, which did not include haptic feedback. In the lifting phase graph on the left, the palm angle fluctuates irregularly between 3 and 5 seconds, with multiple local peaks and valleys. There is no clear, sustained increase or decrease, indicating a less consistent rotational pattern. In the regrasping phase, shown in the right graph, the palm angle initially rises, peaking at approximately 7.4 seconds, before declining sharply. After 8.2 seconds, the angle continues to dip, reaching a minimum before slightly recovering near the end of the time window.

Figure 14 displays the average regrasping angles , measured in degrees, across eight different force directions for both haptic and non-haptic conditions. The x-axis represents the direction from which the simulated slippage force was applied (e.g., "North_North", "South_East", etc.), while the y-axis indicates the corresponding regrasping angle recorded during user trials.

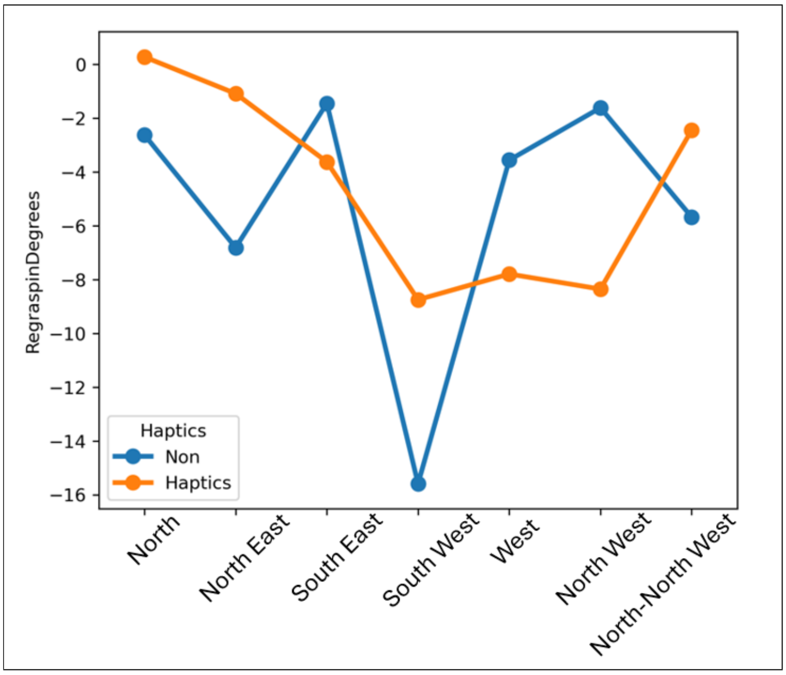

Figure 14: Regrasping Angle in Degrees

For all directions, both haptic and non-haptic trials show negative angle values, indicating rotation in the same general direction. However, the magnitude of these angle changes differs between the two conditions. The non-haptic data (plotted in blue) show greater variability, with a particularly steep dip at the South_West direction, reaching a value near -16 degrees, the largest absolute regrasp angle in the dataset. In contrast, the haptic condition (plotted in orange) shows a more consistent pattern, with smaller magnitude changes. The steepest angle in the haptic condition occurs also at South_West, but it is notably shallower compared to the non-haptic counterpart.

Across directions such as West_West and North_West, the gap between haptic and non-haptic conditions narrows, with both lines approaching similar values. In other directions, particularly North_East and South_East, the haptic responses remain more tightly grouped, whereas the non-haptic responses continue to fluctuate more widely.

4.2 NASA TLX Load Index Results

This section outlines the results obtained from the NASA TLX Load Index that the participants answered. Each participant answered this index every 5 trials, resulting in 4 sets of data depending on haptics and non-haptics and phases of trials. The first phase consisted of trials 0 to 4, the second phase consisted of trials 5 to 9, the third phase consisted of trials 10 to 14, and the final phase consisted of trials 15-19.

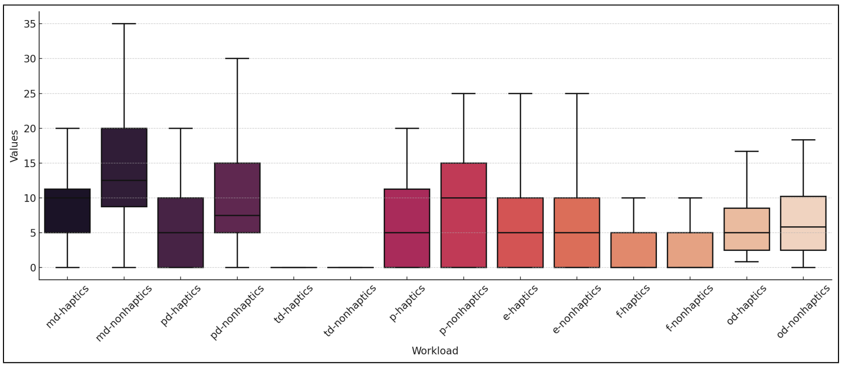

Figure 15: MD = Mental Demand, PD = Physical Demand, TD = Temporal Demand, P = Performance, E = Effort, F = Frustration, OD = Overall Demand

In **Figure 15** we can see a box plot graph showing the index scored of the NASA TLX Load Index sections, which are mental demand, physical demand, temporal demand, performance, effort, and frustration. These sections are split into the haptics and non-haptics results from the index that the participants had filled out. Overall Demand was also included in this graph.

Mental Demand is shown to have a range of 5 to 11 with 2 for the haptics trials, and a range of 9 to 20 for the non-haptics trials. Physical Demand is shown to have a range of 0 to 10 for the haptics trials, and a range of 5 to 15 in the non-haptics trials. Temporal demand for both the haptics trials and the non-haptics trials are limited to a singular value of 1. Performance with haptics is shown to have a range of 0 to 11 while in the non-haptics trials it has a range of 0 to 15. Effort is shown to have a range of 0 to 10 for the haptics trials as well as the non-haptics trials. Frustration is shown to have a range of 0 to 5 for the haptics trials as well as the non-haptics trials. The overall demand for the haptics trials is shown to have a range of 3 to 14 while the overall demand for the non-haptics trials is shown to have a range of 3 to 10.

Sum of Index	Column Labels				
Row Labels	1	2	3	4	Grand Total
Haptics	95.0	34.2	41.7	28.3	199.2
Non	40.0	100.0	20.0	49.2	209.2
Grand Total	**135.0**	**134.2**	**61.7**	**77.5**	**408.3**

Figure 16: NASA-TLX

In the table above, **Figure 16**, we can see the overall index scores from the haptics and non-haptics trials, each phase representing a block of 5 trials as mentioned before. Each phase would have an index score out of 100, a higher index score indicating a higher cognitive load. In phase 1, 2, 3 and 4 of the haptics trials the index scores are respectively 95, 34.2, 41,7 and 28.3. For non-haptics trials it is respectively 40, 100, 20, and 49.2.

5 Discussion

The findings of this study provide convincing insight into the role of haptic feedback in improving users' ability to detect and respond to slippage during virtual object manipulation. The inclusion of fingertip-mounted haptic motors appears to significantly influence both user performance and perceived workload, particularly once participants acclimate to the tactile stimuli.

5.1 Interpretation of Results

Figure 12 shows the angle change from a participant when undergoing a trial that included haptic feedback. From the 3 second mark to the 3.8 second mark, that is the phase in which the participant had lifted the target object and then after that is when the participant dropped it. In the next graph, we can see the regrasping phase of the trial where we can see a progressive increase in the palm angle during this phase. The participant smoothly adjusted their grasping angle, with the angle steadily increasing from 0.1 to 0.35 yRot, implying a more confident and directed response to the haptic feedback triggered during the previous event. The angle change represented in this graph is also in the right direction, we know this because of the positive value in yRot.

By contrast, **Figure 13** represents the non-haptic trial of the same participant. During the regrasping phase, there is a notable change in the palm angle, indicating uncertainty and hesitation when the participant tried to regrasp. This can be attributed to the lack of haptic feedback present for the participant. We can also see that the angle change represented in the graph is a negative yRot value, which is in the wrong direction, as the force originated from the North East.

In **Figure 14**, we can observe the different force directions along with the average regrasping angle of participants during both haptic and non-haptic trials. In the *North* direction, the haptic trials show no angle change, which aligns with our prediction, as participants would not need to rotate their hand during regrasping if the slip direction was directly in front of them. In contrast, the non-haptic trials show a regrasping angle of approximately -2.5 degrees. This suggests that without haptic feedback and relying solely on visual cues, participants were less successful in identifying the direction of slippage. This trend continues throughout the rest of the directions shown in the graph. The haptic trial data displays a relatively stable and consistent downward trend, indicating participants were generally able to detect and respond to the force direction correctly. On the other hand, the non-haptic data is more staggered and inconsistent, reflecting a less accurate estimation of force origin. Most haptic trial regrasp angles appear to be successful across the directions, with a few exceptions. For example, the angle change in the *South East* direction is lower than expected. Participants should have rotated their hands further to the right to match the applied force, but this is not reflected in the data. Meanwhile, the angle in the *North East* direction is appropriate, as only a slight regrasp to the right would be required. The *South West* direction also aligns well, as it involves the largest required angle change among the southward directions. The *West* direction fits the predictions as well, showing the second largest angle change after the southern directions. However, the result for *North West* does not align as expected, it shows a larger regrasp angle than *West*, even though a smaller change would have been anticipated. Lastly, the angle in the *North-North West* direction is consistent with expectations, falling between *North West* and *North*, suggesting a moderate level of rotation as predicted.

In contrast, the non-haptic trial data presents a more irregular and less predictable trend across the various force directions. Without the benefit of haptic feedback, participants relied solely on visual feedback and prior expectation, which appears to have led to greater variability in their regrasping responses. In the directions *South West* and *North East*, the regrasping angles deviate significantly from what would be expected if the force direction had been correctly identified. The sharp dip at the *South West* point, which reaches nearly -16 degrees, may indicate an overcompensation due to uncertainty or delayed recognition of object movement. Conversely, in the directions *South East*, *North West* and *West*, the angle changes are less pronounced than in the haptic trials, which may suggest undercompensation or hesitation in the absence of the haptic feedback cues. This inconsistency in directional response reflects the added challenge of determining slippage purely through visual feedback. Participants were not only less accurate but also less consistent, as seen in the abrupt rises and falls between successive directions. The lack of a stable downward trend, compared to the haptic trials, further emphasizes the difficulty users faced in gauging the direction and angle of the required regrasp. These fluctuations support the observation that tactile input helps ground users' perceptions and enables more informed and deliberate hand adjustments. Overall, the non-haptic results highlight the inherent limitations of visual-only feedback in tasks involving directional slippage.

In **Figure 15**, we observe the NASA TLX (Task Load Index) index scores reported by participants, categorized by whether the trials involved haptic feedback or not. The results consistently indicate that, across nearly all the different indexes, Mental Demand (MD), Physical Demand (PD), Temporal Demand (TD), Performance (P), Effort (E), and Overall Demand (OD), the trials involving haptic feedback resulted in a lower cognitive load. This supports our prediction that the inclusion of haptic feedback reduces cognitive and physical load during the task by providing users with more intuitive, real-time tactile cues to aid in the experimental procedure tasks.

The reduced scores in Mental and Physical Demand suggest that participants found it easier to process and respond to slippage events with the assistance of the haptic system. Similarly, lower Effort scores imply that participants were able to complete the task more efficiently, requiring less perceived exertion. Importantly, the lower Overall Demand score reinforces the effectiveness of the haptic interface in alleviating general task load.

The only notable exception is the Frustration (F) index, which remained unchanged between haptic and non-haptic conditions. This anomaly may be attributed not to the haptic system itself but rather to external factors such as procedural inconsistencies, technical difficulties, or unfamiliarity with the virtual reality environment. Participants may have experienced confusion or discomfort that was not related to the feedback mechanism, leading to frustration that was consistent between both conditions.

In general, the data suggest that haptic feedback plays a significant role in reducing cognitive and physical workload, enhancing the usability and intuitiveness of the VR system for slippage simulation.

5.2 Comparison to Existing Literature

The findings of this study align with the studies that were performed in the literature that were previously reviewed in Chapter 2 of this paper and emphasize the value of haptic feedback in improving user interaction and performance within virtual environments. Previous work has shown that tactile cues can significantly enhance and improve user motor responses and cognitive processing during tasks that involve the manipulation and precision of objects. Looking back at when van der Meijden and Schijven [1] found that integrating haptic feedback in surgical VR simulations improved performance by enabling users to better perceive subtle changes in resistance or force application. This is consistent with the present study, in which participants receiving haptic feedback exhibited more directed and stable regrasping motions during trials involving slippage events.

Similarly, Vapenstad et al. [2] emphasized the role of system responsiveness and high-fidelity feedback in helping user interpretation of tactile signals. Our results echo these insights, as participants in haptic-enabled trials demonstrated angle changes that were not only more consistent in magnitude but also more frequently aligned with the correct direction of the applied force. This suggests a clear advantage in perceptual accuracy when tactile feedback is provided, confirming that the role of haptics in interpreting complex dynamic interactions in VR is vital.

Moreover, previous work by Yang et al. [5] and Otaduy et al. [3] has shown that realistic rendering of contact events such as slip, compliance, and surface friction plays a critical role in immersive environments. In our study, the use of the four fingertip-mounted haptic motors to simulate directional slippage produced measurable improvements in regrasping behavior, supporting the notion that even simplified forms of tactile simulation, such as vibrotactile feedback, can effectively guide motor responses.

The reduced variability in angle changes among participants in haptic trials is also in line with findings from Kovanur Sampath et al. [10], who emphasized the role of tactile feedback in promoting motor-sensory learning and reducing cognitive workload. This is further reinforced by the NASA TLX results in Section 4.2, which showed lower perceived mental and physical demand scores under haptic conditions. Collectively, these outcomes support the hypothesis that haptic feedback not only improves objective task performance but also eases subjective cognitive load, both of which have been frequently cited in the literature as key benefits of tactile augmentation in VR systems.

In summary, the results of this study are strongly aligned with existing research and extend the current understanding of how slippage-specific haptic feedback can be implemented to improve performance and perception in virtual environments. This coincides with the prediction made in Section 1.4 that haptics are a vital component in achieving immersive and responsive human-computer interaction.

5.3 User Adaptation and Learning Curve

Throughout the course of the user trials, a noticeable trend was observed in the behavior of the participants, indicating the adaptation to the haptic feedback system. Although the initial orientation and calibration phase provided users with a basic understanding of the haptic feedback cues, it was evident that repeated exposure to slippage events and motor activation was evident to contribute to improved responsiveness and more consistent regrasping performance over time.

In earlier trial blocks, particularly those involving the first few haptic feedback conditions, several participants exhibited delayed or inconsistent reactions to simulated slippage. This included abrupt or overshooting angle adjustments and slower regrasping actions. However, in subsequent blocks, participants began to demonstrate smoother and more deliberate movements in response to the same stimuli. This behavioral change suggests that users were developing an internal mapping between the direction of the haptic signal and the appropriate corrective motor action.

The learning effect was particularly evident in the reduction of excessive or misdirected hand rotations, which were more prevalent in the initial trials. By the final set of haptic-enabled trials, participants often made faster and more precise regrasping attempts, with angle changes more frequently aligning with the intended direction of slippage. This improvement in control accuracy is further supported by the reduced variability observed in angle magnitudes across later haptic trials.

Interestingly, some degree of adaptation to cross-condition was also observed. Participants who began with haptic trials and transitioned to non-haptic blocks appeared to retain some of the directional awareness and regrasping strategies learned earlier, despite the lack of tactile cues. This suggests that even limited exposure to haptic feedback may contribute to improved motor planning and task familiarity, which can persist even when the feedback is removed.

In general, the observed adaptation indicates that users not only benefited from the inclusion of haptic feedback but also learned to better interpret and utilize it over time. These findings underscore the importance of considering user learning curves in the design of haptic-enabled VR systems, particularly for applications that involve repetitive or high-precision interactions such as remote control, virtual training, or rehabilitation.

5.4 Technical and Experimental Limitations

While the results of this study provide valuable insight into the role of haptic feedback in enhancing slip perception within virtual environments, several technical and experimental limitations must be acknowledged. These limitations may have influenced participant performance and generalization of the findings.

First, the haptic system relied on four vibration motors mounted to the fingertips, each delivering directional slippage haptic feedback signals through rotational motion. Although they were effective in conveying basic directional feedback, the motors do not take into account pressure variation and only simulate slippage in one direction. This limited the realism of the simulated slippage, particularly in cases where subtle differences in slip intensity could have enhanced participant perception. Additionally, the haptic feedback motors introduced minor delays in activation due to serial communication and motor spin-up time, potentially affecting the immediacy of the feedback and its integration with visual stimuli. In order to improve this, the motors can be redesigned to spin in the x, y, and z directions using a sphere to rotate in the direction of the slippage rather than just a wheel that rotates in one direction.

Secondly, the Unity environment and object interaction logic, while functionally robust, operated on a fixed event-based system. This meant that slippage feedback was only triggered at predetermined moments, rather than dynamically in response to grip force or real-time object movement. As a result, the simulated slippage did not adapt to user behavior, which may have reduced the realism of the user study compared to real-world manipulation tasks.

Another limitation was the relatively small sample size and the limited diversity between the participants. Although the study gathered usable data across multiple trials and force directions, the number of unique participants was not large enough to account for a larger variability in hand dynamic and perception thresholds. This constrains the statistical power of the findings and suggests that the results should be interpreted as preliminary rather than definitive.

Lastly, the study design did not incorporate a full counterbalancing of the trial order. In some cases, participants completed haptic trials before non-haptic trials, potentially introducing learning effects or response bias. Although efforts were made to alternate conditions and randomize directions, future studies could benefit from stricter counterbalancing or a crossover design within the subject to minimize order effects.

Despite these limitations, the system performed reliably across all sessions and provided clear contrasts between haptic and non-haptic performance. These insights serve as a strong foundation for refining the feedback mechanism and experimental framework in future work.

5.5 Future Work

While this study demonstrates the potential of haptic feedback in conveying directional slippage within virtual environments, there are several promising directions for future research and system refinement.

One clear direction is to enhance the fidelity of the haptic feedback system. The current implementation uses vibration motors with rotating wheels to simulate slippage in a directional manner, which, while effective, lacks nuance in pressure, surface texture, and continuous force dynamics. Future designs could incorporate more sophisticated actuators, such as linear resonant actuators (LRAs), or force-feedback mechanisms such as servomotors, to simulate not only the direction of slippage, but also its magnitude and resistance. This would enable a more realistic rendering of object interaction, especially for tasks requiring variable grip force or multidirectional slip detection.

Another improvement would be the development of a dynamic feedback model that responds in real time to user behavior. Currently, feedback is triggered at fixed intervals tied to object events in Unity. Future systems could integrate grip force sensors or finger position tracking to dynamically detect onset of slippage and deliver feedback that adapts based on actual slippage velocity or direction. This real-time interaction would more closely replicate the natural touch-based corrections seen in physical environments.

Expanding the system to support multifinger feedback or full-hand interaction is also a possible next step. The current design focuses on the directional signals that are delivered to individual fingertips. However, many natural slippage events involve coordinated responses across multiple fingers or palm regions. A full-hand glove with distributed haptic actuators could provide a richer and more immersive tactile experience, allowing users to interpret spatially complex haptic feedback cues and adjust their grip accordingly.

From an experimental point of view, future studies should involve a larger and more diverse group of participants to validate the findings in a wider range of user profiles. Furthermore, including left-handed participants, users with varying levels of VR experience, or those with motor impairments could uncover insights into the adaptability and inclusivity of the system.

Finally, integrating machine learning to personalize haptic responses based on individual user behavior and performance is another possibility. A feedback system that learns how a specific user responds to slippage cues and adapts the timing or strength of signals accordingly could significantly improve responsiveness and reduce cognitive load.

In summary, future work should focus on improving the precision, responsiveness, and realism of haptic feedback, while also expanding the scope and adaptability of the system to better serve a wider range of users and interaction types.

6 Conclusion

This study investigated the impact of directional haptic feedback on users' ability to perceive and respond to slippage events in a virtual reality environment. Through the integration of a custom-built system involving fingertip-mounted haptic motors, a Unity-based VR application, and a controlled experimental procedure, we were able to compare user performance under haptic and non-haptic conditions.

The results demonstrated that haptic feedback contributed to more consistent and directionally accurate regrasping responses, aligning more closely with the simulated force vectors. Participants who received haptic feedback showed reduced cognitive load, as reflected in lower NASA TLX scores, and exhibited less variability in hand angle adjustments in the trials. These findings support the hypothesis that haptic feedback can improve both the perception of slippage and the precision of corrective motor actions during object manipulation in VR.

Although some limitations were identified, such as the fixed nature of event-based feedback and a relatively small participant pool, the results provide a strong foundation for future research. Further improvements in the haptic feedback motor design and dynamic feedback systems could lead to even more realistic and responsive VR experiences. Ultimately, this work highlights the potential of haptic feedback in the advancement of human-computer interaction in applications such as remote robotics, surgical simulation, and immersive training.

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